Additional file 6. The two proteomic cohorts, and what the matched permutation matched

The main-text section *Protein-level corroboration in two patient-independent datasets* uses two proteomic cohorts. They are not interchangeable, and the differences matter for how far the pooled result can be read. This table records what each source publication and repository record states, and marks what they leave unstated rather than filling it in by inference.

Attribute	**SOMAscan cohort**	**Mass-spectrometry cohort**
Repository / accession	Mendeley Data 83k89shdx5	PRIDE PXD041884
Source publication	Hirohama 2023 (JASN)	Schwab 2024 (Proteomics Clin Appl)
Measurement principle	SomaScan aptamer panel, 1,305 proteins	Label-free LC-MS/MS, Q-TOF, shotgun
Case group as defined by the source	23 individuals with diabetic kidney disease	5 clinically validated diabetic cases, all with nodular sclerosis
Control group as defined by the source	10 healthy controls	7 non-diabetic control cases
Sample labels in the released matrix	Con_01..Con_10, DKD_01..DKD_23	C1, C2d, C3, C4, C6, C7, C9 / D1, D3, D4, D5, D6
Tissue	Kidney cortex	Kidney FFPE tissue curls, over 60% cortex per case
Procurement route stated for the cases	Kidney tissue; route not stated in the accessible text	Explant tissue, archival FFPE, one institution
Procurement route stated for the controls	Kidney tissue; route not stated in the accessible text	Explant tissue, archival FFPE, same institution
Biopsy-versus-nephrectomy asymmetry by design	Cannot be established from the accessible text	Absent: both arms are archival explant cases from one source
Pre-analytical exposure reported	No ischaemia, fixation or storage time reported	No ischaemia, fixation or storage time reported
Already-published protein finding in this cohort	MMP7 is the source publication's headline biomarker	None of our candidates is a named finding of the source

Three things follow. First, the control arms differ – healthy participants in one cohort, non-diabetic pathology cases in the other – so the pooled contrast is diabetic versus non-diabetic rather than one shared case/control definition. Second, the mass-spectrometry cohort draws both arms as archival explant cases from a single institution, which removes the biopsy-versus-nephrectomy contrast from its design; it does not establish that pre-analytical exposure was matched, because neither source reports ischaemia, fixation or storage time. Third, *MMP7* is the headline finding of the publication that produced the SOMAscan cohort, so its significance there restates a published result on the same data. The independent observation is its replication in the mass-spectrometry cohort.

Matched permutation

The pooled enrichment could arise if the candidates were simply abundant, readily detected proteins. To test that, the permutation was repeated so that each null protein was drawn from the background matched to the candidate it replaced: on abundance rank within a caliper of 10 per cent of the background, and, in the mass-spectrometry cohort, on peptide count within one doubling. Draws are without replacement within each permuted set and never cross strata.

Cohort	Candidates	Eligible background	Abundance, candidates	Abundance, matched null
SOMAscan	7	170	10.42	10.5
LC-MS/MS	7	175	21.79	21.8

The unmatched permutation gives $p = 0.035$ and the matched permutation $p = 0.039$ over 200,000 draws, against a null expectation of 2.7 significant candidates unmatched and 2.8 matched. The conclusion does not depend on the candidates being easier to detect than the proteins they are compared with.

Corroboration is not discovery

The pooled result above says that a candidate list built from transcriptomes holds up in protein data. It does not say that the protein data could have produced such a list on its own, and the two need different amounts of evidence: corroboration starts from a list and needs only the few genes that happen to be measured, whereas discovery has to build one and needs a wide search space across several cohorts. We tested the second directly, using the same two datasets and no transcriptomic information, by asking which of the 170 genes both of them quantify reach $q < 0.05$ in both with a concordant direction.

Gene	g (SOMAscan)	q (SOMAscan)	g (LC-MS/MS)	q (LC-MS/MS)	In candidate list
ANXA1	+1.71	0.010	+3.90	0.011	no
GPNMB	+1.91	0.008	+3.01	0.024	no
GZMA	+1.10	0.032	+2.57	0.029	no
MMP7	+2.04	0.008	+3.34	0.045	yes

Four genes qualify, three of them outside the transcriptomic candidate list. That is not more than chance: under independence 2.1 are expected, and a permutation that shuffles the gene labels of one dataset while holding each dataset to its own distribution of significance and direction gives $p = 0.10$ over 200,000 draws (binomial $p = 0.15$). The protein layer here has 2 datasets, 45 patients and 170 shared genes, against 7 cohorts, 258 patients and 9,900 genes on the transcriptome side; with two datasets a leave-one-dataset-out procedure degenerates to training on one and scoring on the other. We therefore report the protein data as corroboration only, and record this negative result rather than leaving the stronger reading available.

Did we measure the protein of that gene

Neither platform names a gene directly. An aptamer is assumed to bind its target, and mass spectrometry assigns peptides to proteins, some of which belong to more than one. The table below records, for every candidate that either cohort measured, the aptamer identifier, the accession, and how many distinct peptides are unique to that protein rather than shared with others in its group.

Gene	Aptamer	Accession	Unique	Shared	Shared with
ACTN1	–	P12814	8	17	O43707 (ACTN4), P35609 (ACTN2), Q08043 (ACTN3)
CA10	seq_3289_19_2	–	0	0	–
COL1A2	–	P08123	7	0	–
CTNNBIP1	–	Q9NSA3	1	0	–
LUM	seq_13114_50_3	P51884	4	0	–
MMP2	seq_4160_49_1	–	0	0	–
MMP7	seq_2789_26_2	P09237	2	0	–
RNASE6	–	Q93091	1	0	–
THBS2	seq_3339_33_1	–	0	0	–
TNFRSF12A	seq_5138_50_3	–	0	0	–
TNNT2	seq_5315_22_3	–	0	0	–
VCAN	–	P13611	3	0	–

Two entries carry caveats and both are genes the mass-spectrometry cohort called significant. *ACTN1* has 8 unique peptides out of 25, the rest shared with other actinins including the podocyte gene *ACTN4*; its value cannot be attributed to *ACTN1* alone. *MMP7* has an unambiguous accession but only 2 distinct peptides. Both are reported rather than dropped.

The candidate list was fixed before these results

The manuscript states that the analysis plan was fixed on the first cohort and applied unchanged. Timestamps cannot demonstrate that, because they change on every re-run. What can be shown is the direction of dependency: the stage that produces the candidate table takes the corrected-selector output, the literature counts, the KPMP evidence and the per-diagnosis pattern as inputs, and none of the proteomic results among them. Every proteomic stage runs after it. Verdict: `frozen`. The table analysed here has SHA-256 `4ec701bce71b4ec5b4ba83558a47efbd...` and its input `candidates_gated.tsv` has SHA-256 `0ddfe629a4c02bcac77775f180b8a6be...`.

What each novelty tier rests on

Reference verification confirms that a cited paper exists; it does not confirm the claim that a candidate is sparsely represented in the diabetic kidney disease literature. That claim comes from PubMed counts, so the queries, the date they were run and the returned identifiers are recorded per candidate in `literature_ledger.tsv`. Counts grow over time, so the frozen values the manuscript reports were re-queried on 2026-09-02: no candidate changed tier. Four symbols

collide with common non-gene abbreviations and are flagged as unreliable rather than counted, with the reason given for each.